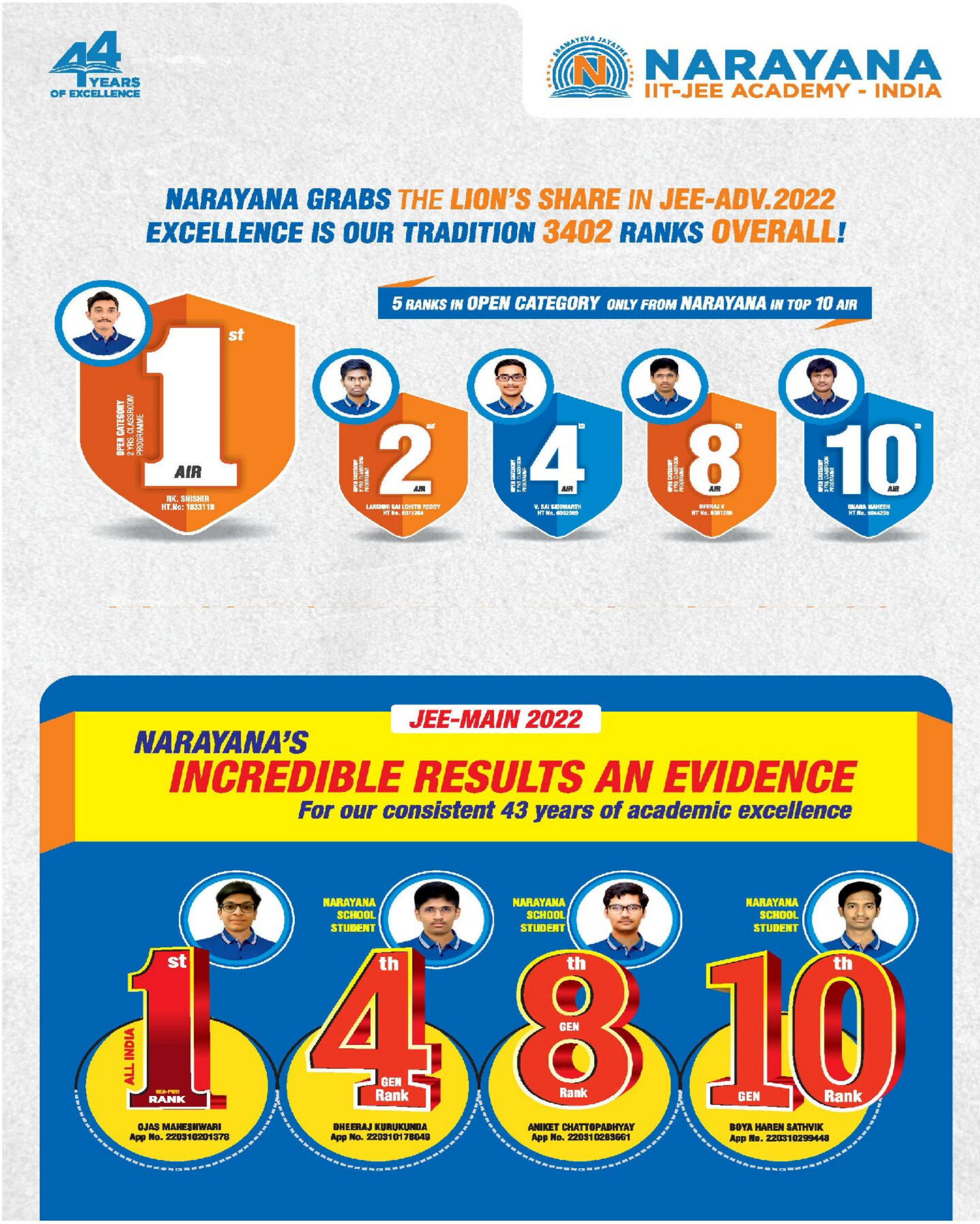


CHEMISTRY

PHYSICAL CHEMISTRY PART-II(SR)

JEE ADVANCED IMPORTANT QUESTIONS

Single correct questions

1. For the given reaction step I is endothermic and step II is exothermic such that overall reaction is exothermic

$$A_2(g)+B_2(g)\underset{K_2}{\overset{K_1}{\rightleftharpoons}}3C(g)\underset{K4}{\overset{K_3}{\rightleftharpoons}}D(g)$$
step I Step II

Which of the following statement is not true?

A) Step I is favored by low P, high T whereas step II is favored by high P and low T.

B) Equilibrium constant for step I increase with temperature whereas equilibrium constant for step II decreases with temperature

C) Rate constant for both the steps increase with temperature.

D) The ratio of two equilibrium constants for step I and step II respectively is $\frac{[D]}{[A_2][B_2]}$.

Ans: D

Sol:

Follow Le Chatelier principle for (a) and for (b) choices

$$2.303\log\frac{K_{C_2}}{K_{C_1}}=\frac{\Delta H}{R}\left[\frac{T_2-T_1}{T_1T_2}\right]$$

and $K = Ae^{-E_a/RT}$ for (c)

Also, $K_{C_1}=\frac{[C]^3}{[A_2][B_2]}$

and $K_{C_2}=\frac{[D]}{[C]^3}$

$$\therefore \quad \frac{K_{C_1}}{K_{C_2}}=\frac{[C]^6}{[A_2][B_2][D]}$$

2. The ratio of initial rate and the rate at time t for an elementary reaction:

$2A+B\rightarrow P$,

When $[A]_0=0.4M$ and $[B]_0=0.3M$ as well as $[A]_t=0.2M$ and $[B]_t=0.2M$ is

A) 0.6 B) 0.5 C) 0.8 D) 0.2

Ans:- A

Sol:-

$$r_0=K[A]_0^2[B]_0^1=K[0.4]^2[0.3]^1=4.8\times10^{-2}$$

$$r_t=K[A]_t^2[B]_t^1=K[0.2]^2[0.2]^1=0.8\times10^{-2}$$

$$\therefore \quad \frac{r_0}{r_t}=\frac{4.8\times10^{-2}}{0.8\times10^{-2}}=\mathbf{0.6}$$

3. Two solutions of $1N\,Na_2SO_4$ and 1N NaCl having osmotic pressure π_1 and π_2 respectively at 300K are separated through semipermeable membrane. Which statements are correct for them if both are 100% ionized?

i) $\pi_1<\pi_2$

ii) $1N\,Na_2SO_4$ is hypertonic solution

iii) 1N NaCl is hypertonic solution

iv) $\pi_1 > \pi_2$

v) water moves from Nacl solution to Na_2SO_4 solution

A) (i) ,(iii) B) (ii),(iv) C) (ii),(v) d) (i),(ii)

Key:- A

Sol:-

$$1N\ Na_2SO_4 = 0.5M\ Na_2SO_4$$

$$1N\ NaCl = 1M\ NaCl$$

$$\therefore \quad \pi_1 = \pi_{Na_2SO_4} = 0.5 \times S \times T \times 3 = \frac{3}{2}ST$$

$$\pi_2 = \pi_{NaCl} = 1 \times S \times T \times 2 = 2ST \text{ (Hypertonic)}$$

4. 1 g of a complex $\left[Cr(H_2O)_5 Cl\right]Cl_2 .\ H_2O$ (Molar mass 266.5) was passed through a cation exchanger to produce HCl. The acid liberated was diluted to 1 litre. The normality of this acid solution is

A) 5×10^{-3}N B) 7.5×10^{-3}N C) 7.5×10^{-2}N D) 7.5×10^{-1}N

Key:- B

Sol:- Mole of complex $= \dfrac{1}{266.5}$

1 mole of complex produces 2 mole HCl

$$\left[Cr(H_2O)_5 Cl\right]Cl_2 \rightarrow \left[Cr(H_2O)_5 Cl\right]^{2+} + 2Cl^-$$

$\therefore$ mole of HCl formed $= \dfrac{1}{266.5} \times 2$

$\therefore$ $N_{HCl} = \dfrac{2}{266.5\times1} = 7.5\times10^{-3}$

5. A certain reaction proceeds in a sequence of three elementary steps with the rate constants K_1, K_2 and K_3. If the observed rate constant (K_{obs}) of the reaction is expressed as $K_{obs} = \left[\dfrac{K_1}{K_2}\right]^{1/2} \times K_3$; the observed energy of activation of the reaction is.

A) $\dfrac{1}{2}\left[\dfrac{E_1}{E_2}\right] + E_3$ B) $\dfrac{E_3 + E_1}{2}$ C) $E_3\left[\dfrac{E_1}{E_2}\right]^{1/2}$ D) $E_3 + \dfrac{1}{2}\left[E_1 - E_2\right]$

Ans:- D

Sol:-

$$K_{obs} = K_3\left[\frac{K_1}{K_2}\right]^{1/2}$$

$$= A_3 e^{-E_3/RT} \times \left[\frac{A_1 e^{-E_1/RT}}{A_2 e^{-E_2/RT}}\right]^{1/2}$$

$$\therefore \quad K_{obs} = \frac{A_3\sqrt{A_1}}{\sqrt{A_2}} \times e^{-\left[\frac{E_3}{RT} + \frac{E_1 - E_2}{2RT}\right]}$$

$$K_{obs} = A \cdot e^{-E_{obs}/RT}$$

$$\therefore \quad E_{obs} = E_3 + \left[\frac{E_1}{2} - \frac{E_2}{2}\right]$$

6. For the reaction $A + B \rightleftharpoons X^{+} \rightarrow P;\ E_a = 20\,kJ\,mol^{-1}$ at 300 K. The enthalpy change for the formation of activated complex from the reactants in kJ mol^{-1} is .

A) 12 B) 15 C) 23 D) 24

Key:- B

Sol:-

$$\Delta H^\circ = E_a - \Delta V(g) \times RT$$
$$= 20 - 2 \times 8.314 \times 10^{-3} \times 300 = 20 - 5 = 15\ \mathbf{kJ\ mol^{-1}}$$

7. Select the incorrect statement

A) The degree of polymerisation does not affect the percent composition in monomer and polymer

B) ^{12}C is the standard used for all atomic, molecular and formula masses

C) Avogadro's number does not depend on the basis of the atomic weight scale

D) For every dilute solution, mole fraction of solute $= \dfrac{\text{molality} \times \text{molar mass of solvent}}{1000}$

Ans:- C

Sol:- Avogadro's number obtained by ^{16}O and ^{12}C standard are different and slightly differ and in 1961 it was decided to base on ^{12}C.

8. Select the correct statements:

1. Uncharged atoms or molecules never crystallize in simple cubic structures

2. An end centred unit cell cannot be cubic

3. The number of octahedral sites per sphere in a fcc structure is one

4. A closed container of water at 100^0C cools faster in comparison to open container of water at 100^0C

A) 1,2,3,4 B) 1,2,4 C) 1,3,4 D) 1,2,3

Key:- D

Sol:- The water cools faster in open container as it cal also evaporate

9. Which one is incorrect for the ideal solution of two volatile liquid when n mole of A and N mole of B are mixed; P'_a and P'_b are partial pressures of A and B in mixture and P_T is total pressure?

A) The number of moles of each substance in the vapour phase is proportional to its partial pressure in vapour phase

B) $P'_a = P_T \cdot X_{a(vap)}$ and $P'_b = P_T \cdot X_{b(vap)}$

C) $X_{a(vap)} = \frac{P'_a}{P_T}$

D) $\frac{P'_a}{P'_b} = \frac{w_b}{m_b} \times \frac{m_a}{w_a}$

Key:- D

Sol:- $P'_a / P'_b = \frac{n_a}{n_b}$

10. E^0 for two reactions are given below

$$Cr^3 + 3e \longrightarrow Cr; E^0 = -0.74V$$

$$OCl^- + H_2O + 2e \longrightarrow Cl^- + 2OH^-; E^0 = 0.94V$$

What will be E^0 for

$3OCl^- + 2Cr + 3H_2O \longrightarrow 2Cr^{3+} + 3Cl^- + 6OH^-$?

A) 1.68 V B) 0.20 V C) -1.68 V D) -0.20 V

Key:- A

Sol:- $E^0 = E^0_{OP_{Cr}} + E^0_{RP_{OCl^-}} = 0.74 + 0.94 = 1.68V$

11. Which property is not correct regarding the given trend for some ions?

$Na^+ < Mg^{2+} < Al^{3+}$

A) E^0_{OP} B) Extent of hydration C) ionic mobility D) None of these

Ans:- A

Sol:- E^0_{OP} order: does not depend upon the size of ion

12. Which will show maximum ΔT_f in 1000 g of 100% sulphuric acid solution?

A) One mole triphenyl methanol B) One mole ethanol

C) One mole methanol D) Same in all

Key:- A

Sol:-

$$(Ph)_3 \cdot C \cdot OH + H_2SO_4 \rightleftharpoons Ph_3COH_2^+ + HSO_4^-$$

$$Ph_3COH_2^+ \rightleftharpoons Ph_3C^+ + H_2O$$

$$H_2SO_4 + H_2O \rightleftharpoons H_3O^+ + HSO_4^-$$

$$\overline{Ph_3COH + 2H_2SO_4 \rightleftharpoons Ph_3C^+ + H_3O^+ + 2HSO_4^-}$$

(four ions)

$$MeOH + H_2SO_4 \rightleftharpoons MeOH_2^+ + HSO_4^-$$ (two ions)

13. Which of the following cells possess ΔG^0 value '-ve' as per the given cell notation?

A) $Zn|Zn^{2+}||Ag^+|Ag$ B) $Pt : H_2|HCl||Ca^{2+}|Ca$

C) $Pt : HCl|Cl_2||H_2|HCl : Pt$ D) $Pt : Cl_2|HCl||HCl|H_2 : Pt$

Key:- A

Sol:-

14. In which of the following cases the clear final solution has higher freezing point than clear original solution?

A) Passing sufficient HCl gas into aq.AgNO3

B) Adding sufficient $HgI_{2(S)}$ to $KI_{(aq)}$

C) Adding sufficient $BaCl_{2(S)}$ solid to $Na_2SO_{4(aq)}$

D) Adding sufficient solid Na_2SO_4 to aqueous $BaCl_2$

Key:- B

Sol:-

15. An ionic solid (Gram formula weight = 60.23 g) is crystallized in rock salt structure with 4 A^0 edge length of unit cell. The observed density of the crystalline solid is 5 gm/cc. How many moles of unit cells would contain 16 moles of the given salt?

A) 5 B) 4 C) 3 D) 2

Key:- A

Sol:-

16. Certain quantity of metal is crystallised 60% in ccp structure 40% in hcp structure. What is the ratio of number of unit cells in ccp and hcp?

A) 3 : 2 B) 1 : 1 C) 9 : 4 D) 4 : 9

Key:- C

Sol:-

17. In an ABABAB…… arrangement of oxide ions, 50% of octahedral voids are occupied by A^{n+} ions and 25% of tetrahedral voids are occupied by B^{y+} ions. Here, A and B may be

A) Ca^{2+}, Na^{+} B) K^{+}, Al^{3+} C) Ca^{2+}, Na^{+} D) Ca^{2+}, Th^{4+}

Key:- B

Sol:-

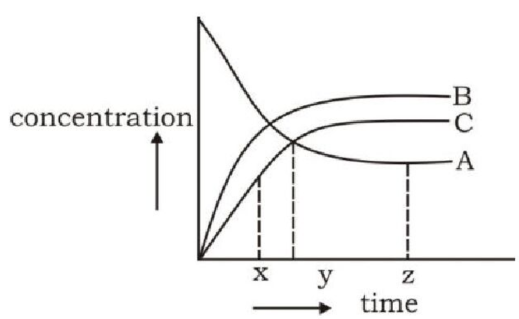
18. In a one molal solution, the solute undergoes dimerisation in an organic solvent. (Freezing point $= 4^0C$ $K_f = 8K\ kg\ mol^{-1}$) Which of the following freezing point of the solution is not acceptable?

A) -4.2^0C B) 1.2^0C C) 2.4^0C D) -3.2^0C

Key:- D

Sol:-

19. $A \rightleftharpoons 2B + C$



Which statement is correct for the reversible reaction given above?

A) At time x, $Q_c > K_c$ B) At time z, $Q_c < K_c$ C) At time y, $Q_c < K_c$ D) At time z, $Q_c > K_c$

Key:- C
Sol:-

20. What are the units of rate constant for decomposition of gaseous NH_3 on 'Pt' surface?

A) $\sec^{-1}$ B) $mol\,L^{-1}\sec^{-1}$ C) $mol^{-1}L\sec^{-1}$ D) $mol^{-2}L^{2}\sec^{-1}$

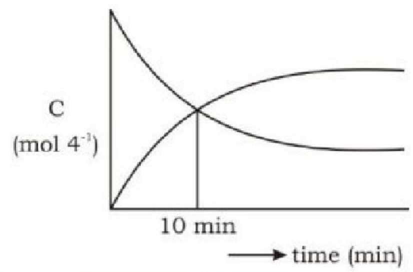
Key:- B
Sol:-

21. 3.95 moles of rock salt produced one mole of unit cells. What is the percentage of missing 'ion pairs' due to Schottky defects?

A) 5% B) 1.25% C) 12.5% D) 22.5%

Key:- B
Sol:-

22. $A \rightarrow \frac{1}{3}B\left(1^{st}\ order\right)$



What is the half-life of the reaction?

A) 10 min B) 5 min C) 20 min D) 8 min

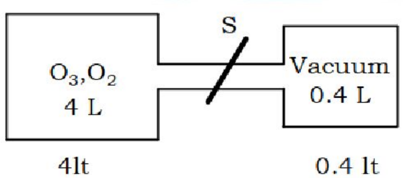
23. Elivation in boiling point of one molal $COCl_3.xNH_3$ solution is equal to ebuliscopic constant of water. Which statement is 'NOT' correct about the given compound?

A) It can show fac-men isomerism B) It's vant Hoft factor is 1.
C) It's solution can not conduct electricity D) It gives white precipitate with $AgNO_3$

Key:- D

Sol:-

24.



Initially, the molecular weight of ozone and oxygen mixture shown above is 44. When the stopper is opened, O_3 is further dissociated to O_2 .But the mixture continued with the same pressure. What is the ratio of moles of O_3 to O_2 in the final mixture?

A) 1 B) 2 C) 3 D) 4

Key:- A
Sol:-

25. The number of unit cells of "MX" to be dissolved in water to prepare 200 ml of saturated solution is $x \times 10^{16}$. Where x is_______(MX adopts rock salt structure and K_{sp} of MX is 10^{-12})

A) 1 B) 2 C) 3 D) 4

Key:- C
Sol:-

26. In Langmuir's model of a gas on a solid surface.

A) The rate of dissociation of adsorbed molecules from the surfave does not depend on the surface covered

B) The adsorption at a single site on the surface may involve multiple molecules at the same time.

C) The mass of gas striking a given area of surafce is proportional to the pressure of the gas.

D) The mass of gas striking a given area surface is independent of the pressure of the gas

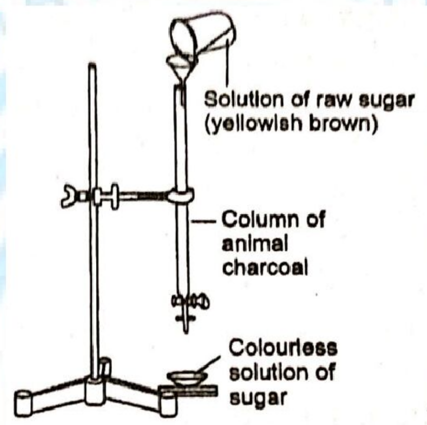
Key:- C

Sol:- Langmuir model shows the dependence of surface coverage of an adsorbed gas on the pressure of gas above the surface at a ffixed temperature.

$$\theta = \frac{kp}{1+kp}$$

Where p is the pressure of the gas and θ is the fraction of adsorbed sites.

27. Which of the following phenomenon is applicable to the process shown in the figure?



A) Absorption B) Adsorption C) Coagulation D) Emulsification

Key:- B

Sol:- $\underset{(Yellowish\ brown)}{Sugar\ solution} \xrightarrow{Charcoal}$ Colorless solution of sugar

i.e., charcoal adsorbed the colour (dye).

28. Which of the following statements are correct?

I. Mixing two oppositely charged sols neutralizes their charges and stabilizes the colloid

II. Presence of equal and similar charges on colloidal particles provides stability to the colloids

III. Any amount of dispersed liquid can be added to emulsion without destabilizing it.

IV. Brownian movement stabilizes sols.

Choose the correct option.

A) I and II B) II and III C) II and IV D) II, III and IV

Key:- C

Sol:- Brownian movement stabilizes sols. Equal and similar charges on colloidal particles provide stability to the colloids.

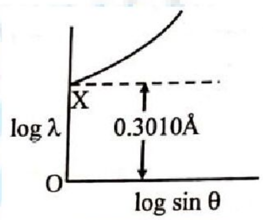
29. Select right expression for determining packing fraction (P.F) of NaCl unit cell (assume ideal), if ions along an edge diagonal are absent:

A) $P.F. = \dfrac{\frac{4}{3}\pi\left(r_+^3 + r_-^3\right)}{16\sqrt{2}r_-^3}$ B) $P.F. = \dfrac{\frac{4}{3}\pi\left(\frac{5}{3}r_+^3 + 4r_-^3\right)}{16\sqrt{2}r_-^3}$

C) $P.F. = \dfrac{\frac{4}{3}\pi\left(\frac{5}{2}r_+^3 + r_-^3\right)}{16\sqrt{2}r_-^3}$ D) $P.F. = \dfrac{\frac{4}{3}\pi\left(\frac{7}{2}r_+^3 + r_-^3\right)}{16\sqrt{2}r_-^3}$

Key:- C

30. For a second order diffraction using X-ray Bragg's equation, for a crystal with separation between two layers as $d\overset{0}{A}$, following graphical study was observed when varying θ at different λ.

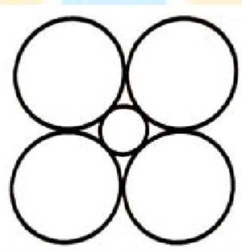


What is the value of d?

A) 0.3010 $\overset{\circ}{A}$ B) 0.6020 $\overset{\circ}{A}$ C) 2.0 $\overset{\circ}{A}$ D) 4.0 $\overset{\circ}{A}$

Key:- A

31. LiCl has NaCl type structure with edge length 514 pm. Assuming that the Li^+ ion is small enough that the chloride ions are in contact, then radius of Cl^{-1} ion is



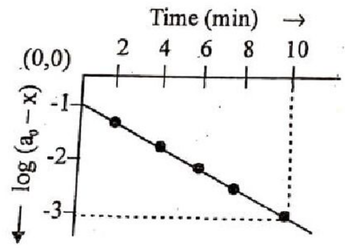
A) 257 pm B) 182 pm C) 364 pm D) 223 pm

Key:- C

32. For the first order decomposition of $SO_2Cl_2(g)$

$SO_2Cl_2(g) \rightarrow SO_2(g) + Cl_2(g)$

A graph of log $(a_0 - x)$ vs t is shown in figure, what is the rate constant $\left(\sec^{-1}\right)$?



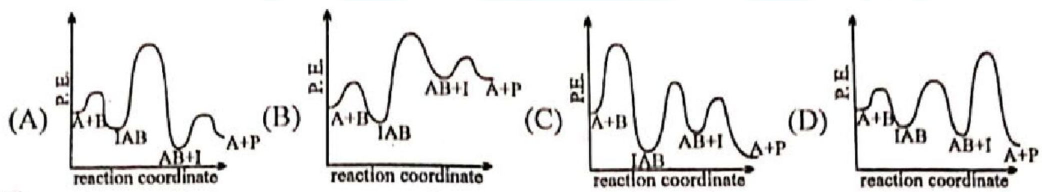
A) 0.2 B) 4.6×10^{-1} C) 7.7×10^{-3} D) 1.15×10^{-2}

Ans:- C

33. The following mechanism has been proposed for the exothermic catalyzed complex reaction.

$$A+B\underset{}{\overset{fast}{\rightleftharpoons}}I\,AB\xrightarrow{k_1}AB+I\xrightarrow{k_2}P+A$$

If k_1 is much smaller than k_2. the most suitable qualitative plot of potential energy (P.E) versus Reaction coordinate for the above reaction.



Ans:- A

34. The following mechanism has been proposed for the reaction of NO with Br_2 to from NoBr. If the second step is RDS. The order of reaction w.r.t. to NO:

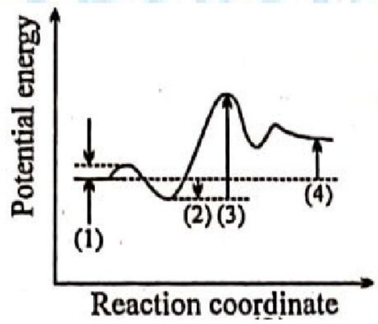
$$NO(g)+Br_2(g)\rightleftharpoons NoBr_2(g)$$

$$NOBr_2(g)+NO(g)\rightarrow 2NoBr(g)$$

A) 2 B) 1 C) 0 D) 3

Key:- A

35. Choose the correct set of identifications.



	(1)	(2)	(3)	(4)
A)	ΔE *for* $E+S\rightarrow ES$	E_a *for* $ES\rightarrow EP$	$\Delta E_{overall}$ *for* $S\rightarrow P$	E_a *for* $EP\rightarrow E+P$

B)	E_a *for* $E+S \to ES$	ΔE *for* $E+S \to ES$	E_a *for* $ES \to EP$	$\Delta E_{Over\ all}$ *for* $S \to P$
C)	E_a *for* $ES \to EP$	E_a *for* $EP \to E+P$	$\Delta E_{overall}$ *for* $S \to P$	ΔE *for* $EP \to E+P$
D)	E_a*for* $E+S \to ES$	E_a *for* $ES \to EP$	E_a *for* $EP \to E+P$	$\Delta E_{overall}$ *for* $S \to P$
E)	ΔE *for* $E+S \to ES$	$\Delta E_{overall}$ *for* $S \to P$	$\Delta E\ for_1$ $Ep \to E^1 + P$	E_a *for* $EP \to E+P$

Key:- B

MORE THAN ONE CORRECT ANSWER

36. Select the correct statements

A) Bragg's equation is valid only if $2d > n\lambda$.

B) Presence of excess Li in LiCl develops pink colour in it.

C) Greater is the number of F-centres, greater is the intensity of colour developed.

D) Conducting power of Si and Ge is increased dramatically if impurities of P or As are incorporated into their lattice.

E) The number of Schottky defect increases with increasing temperature which leads to decrease in density of solid

Key:- A,B,C,D,E

Sol:- All are facts

37. Which of the following statements are correct?

A) Adsorption is spontaneous process with $\Delta G, \Delta H$ and $\Delta S = -ve$.

B) For adsorption both ΔH and ΔS are negative but $T\Delta S < \Delta H$.

C) Both absorption and adsorption cannot occur simultaneously.

D) Silica gel absorbs water.

E) Catalytic action and colloidal nature is based on the phenomenon of adsorption.

Key:- ABE

Sol:-

Silica gel adsorbs H_2O. The phenomenon of simultaneous adsorption and absorption is called sorption.

38. Select the correct statements.

A) Passage of three Faraday of charge through $Mg(NO_3)_2$ solution using Pt electrodes produces 3eq. of Mg at cathode.

B) Passage of two Faraday of charge through AgF(aq.) deposits 1eq. of Ag at cathode and liberates 11.2 litre of fluorine at anode.

C) The calomel electrode is reversible with respect to Cl^-.

D) In electrolyte cell, flow of current and electron is from anode to cathode

E) Mobility of H^+ ion is maximum.

Key:- CE

Sol:-

Mg and F_2 cannot be obtained on electrolysis of aqueous solution.
Only current flows from anode to cathode in electrolyte cell.

39. Select the correct statements about a homogeneous mixture of two liquids.

A) The vapour phase is richer in the component whose addition to the liquid mixture results an increase of total vapour pressure.

B) The liquid phase is richer in the component whose addition to the liquid mixture results in a decrease of total vapour pressure.

C) The vapour phase is richer in the component whose addition to the liquid mixture causes in a decrease in its boiling point.

D) The vapour phase is richer in the component whose addition to the liquid mixture causes in an increase in its boiing point.

Key:- ABC

Sol:-

40. A super saturated solution contains more of the dissolved salt than at equailibrum. The super saturation (a metastable state) occurs during percipitation of a solution. The phenomenon of super saturation can be minimised by:

A) precipitation from dilute solution

B) Adding dilute precipitating reagent with effective stirring

C) precipiration in hot solution

D) precipitation at as low pH as is possible to still maintaining precipitation

Key:- ABCD

Sol:-

41. A definite mass of a non-volatile solute (molar mass m) is dissolved in definite mass of solvent (molar mass M). Which statements are correct?

A) The b.pt. of solution is higher than that of the solvent.

B) the f.pt. of solution is lower than that of the solvent.

C) The depression in freesing point for a solution is greater in magnitude than the elevation in boiling point for same solution.

D) Molal elecation constant of solvent is always higher than its molal cryoscopic costant.

Key:- ABD

Sol:-

42. Which equations correctly represents Clausius-Clapeyron equation?

A) $\frac{dP}{P} = \frac{\Delta H}{R}\frac{dT}{T^2}$

B) In $\frac{P_2}{P_1} = \frac{\Delta H}{R}\frac{[T_2 - T_1]}{T_1T_2}$

C) In $\frac{P}{P^0} = \frac{-\Delta H}{RT} +$ integration constant; Where $P^0 =$ unit vapour pressure

D) In $\frac{P^0}{P} = \frac{-\Delta H}{RT} +$ integration constant; where $P^0 =$ unit vapour pressure

Key:- ABC

Sol:-

43. Which of the following statements are correct, if a solution contains total 1 mole of solute and solvent bothm and the molar masses of solute and solvent being M_1 and M_2 ? X_1 and X_2 are mole fractions of solute and solvent, M and m are molarity and molality of solutions.

A) $M = \frac{X_1.\rho}{(X_1M_1 + X_2M_2)}$ B) $X_1 = \frac{MM_2}{\rho + M(M_2 - M_1)}$

C) $M = \frac{X_1\rho}{M_2}$ for very dilute solution D) $m = \frac{X_1\rho}{M_2}$ for very dilute solution

Key:- ABCD

Sol:-

44. The corect statement(s) partaining to the adsorption of a gas on a solid surface is (are)

A) Adsorption is always exothermic.

B) Physisorption may transform into chemisorptions at high temperature.

C) Physisorption increases with temperature but chemisorptions decreases with increasing temperature.

D) Chemisorption is more exothermic than physisorption; however, it is very slow due to higher energy of activation

Key:- ABD

Sol:-

(a) During adsorption, there is always decrease in surface energy which appears as heat. In other words, we can say that adsorption is an exothermic process.

(b) Physisorption may transform into chemisorptions at hiogh temperature. Fro example, dihydrogen if first adsorbed on Ni by van der Waals forces, molecules of hydrogen then dissociate to form H atoms, which are held on the surface by chemisorptions.

(c) Low temperature is favorable for physisorption; it decreases with increase of temperature. High temperature is favorable for chemisorptions; it increases with increase of temperature.

(d) Energy of adsorption is high in chemisorptions which is about 80 – 240 kJ mol^{-1} as compared of physiosorption where the energy is about 20 – 40kJ mol^{-1}.

45. Choose the correct reason (s) for the stability of the lyophobic colloidal particles.

A) Preferential adsorption of ions on their surface from the solution.

B) Preferential adsorption of solvent on their surface from the solution.

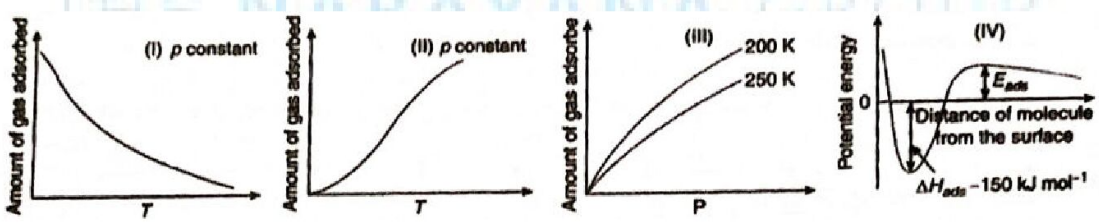
C) Attraction between different particles having opposite charge in their surface.

D) Potential difference between the fixed layer and the diffused of opposite charges around the colloidal particles

Key:- AD

Sol:- In case of lyophobic sols, the ionic colloid adsorbs ions common to its own lattice during the preparation of the sol. So they are stable and also the potential difference which arises between the fixed layer and the diffused layer makes lyophobic sols more stable

46. The given graph/data I,II,III and IV represent general trends observed for different physisorption and chemisorptions processes under mild conditions of temperature and pressure. Which of the following choice (s) about I,II,III and IV is (are) correct?



A) I is physisorption and II is chemisorption

B) I is physisorption and III is chemisorption

C) IV is chemisorptions and II is chemisorption

D) IV is chemisorptions and III is chemisorption

Key:- AC

Sol:-

In case of physisorption, with the increase of temperature and pressure the rate of adsorption decreases because according to Le Chatelier's principle, increases of temperature and pressure will shift the equilibrium to the left

Adsorbate + Adsorbent $\rightleftharpoons$ Adsorption + Heat

This is shown in Graphs I and III; whereas in case of chemisorptions, there is a formation of strong bond between the adsorbate and the adsorbent and so the rate of adsorption increases with increase in temperature (Graphs II and IV).

47. Which of the following are incorrect statements?

A) Hardy schulz rule is related to coagulation

B) Brownian moment and Tyndall effect are the characteristic of colloids.

C) In gel, the liquid is dispersed in liquid

D) Lower the gold number, more is the protective power of lyophillic sols.

Key:- CD

Sol:- In gel, liquid dispesed in solid

48. The origin of charge on colloidal solution is

A) Frictional rubbing

B) Electron capture during Berdig's arc method

C) Selective adsorption of ion on their surface

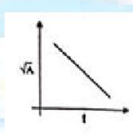
D) It is due to addition of protecitve colloids

Key:- ABC

Sol:-

49. For the reaction $A \rightarrow B$, the rate law expression is $-\frac{d[A]}{dt} = k[A]^{1/2}$. If initial concertation of [A} is $[A]_0$, then

A) the integrated rate expression is $k = \frac{2}{t}\left(A_0^{1/2} - A^{1/2}\right)$

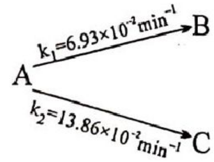
B) The graph of $\sqrt{A}$ Vs t will be 

C) The halftime period $t_{1/2} = \frac{K}{2[A]_0^{1/2}}$

D) The time taken for 75% completion of reaction $t_{3/4} = \frac{\sqrt{[A]_0}}{k}$

Key:- ABD

50. Consider the reaction,



A,B and C all are optically active compound. If optical rotation per unit concertation of A,B and C are $60^0, -72^0, 42^0$ and initial concentration of A is 2M then select write statement (s).

A) Solution will be optically active and dextro after very long time

B) Solution will be optically active and levo after very long time

C) Half life of reaction is 15 min

D) After 75% conversion of A into B and C angle of rotation of solution will be 36^0.

Key:- AD

MATRIX MATCHING

51. Match the List A with List B. One or more match are possible.

List-A	**List-B**
a. Only β-emitter	1. $^{14}_{6}C$
b. Maximum n/p ratio	2. $^{3}_{1}H$
c. Positron emitter	3. $^{11}_{6}C$
d. $\alpha,\beta-$emitter	4. $^{235}_{92}U$
e. $(4n+3)$ series	5. $^{81}_{37}Rb$
f. K-electron capture	6. $^{223}_{87}Fr$

Ans:- a-1,2,6;b-2;c-3;d-4;e-4,6;f-5.

Sol:-

52. match the following

List-A	List-B	List-C
1. Coulometry	a. Electro deposition	i) Analysis of a gas sample
2. Eudiometry	b. Combustion in oxygen	ii) Copper voltameter
3. Potentiometry	c. Titration	iii) Optical rotation
4. Conductometry	d. micellisation study	iv) migration of ions
5. Polarimetry	e. optical activity	v) glass electrode

Ans:- 1-a-ii;2-b-I;3-c-v;4-d-iv;5-e-iii

Sol:-

53. Match the column-I with column-II and select the correct option from the codes given below

Column-I	Column-II
A. Sulphur vapour passed through cold water	1.Normal electrolyte solution
B. Soap mixed with water above critical micelle concetraction	2. Molecular colloids
C. White of egg whipped with water	3. Associated colloids
D. Soap mixed with water below critical micelle concetration	4. Macromolecular colloidal

A) $A \to (4); B \to (2); C \to (1); D \to (3)$ B) $A \to (2); B \to (3); C \to (4); D \to (1)$

C) $A \to (4); B \to (1); C \to (3); D \to (2)$ D) $A \to (1); B \to (2); C \to (4); D \to (3)$

Key:- B

Sol:-

Soap mixed with water above the critical micelle concentration (CMC), then micelles are formed i.e., associated colloids. Soap mixed with water below CMC then, micelle did not form and behaves as normal electrolyte solution.
White of egg (i.e., protein) whipped with water, then macromolecular colloids are formed.

54. Match the column-I with Column-II and select the correct option from the codes given below

Column-I	Column-II
A. Protective colloid	1. $FeCl_3 + NaOH$
B. Liquid-liquid colloid	2. Lyophilic colloids
C. Positively charged colloid	3. Emulsion
D. Negatively charged colloid	$FeCl_3 +$ Hot water

A) $A \to (2); B \to (3); C \to (4); D \to (1)$ B) $A \to (2); B \to (3); C \to (1); D \to (4)$

C) $A \to (2); B \to (1); C \to (4); D \to (3)$ D) $A \to (1); B \to (2); C \to (3); D \to (4)$

Key:- A

Sol:-

Lyophilic colloids are called protective colloids.

Liquid: Liquid colloids are known as emulsion

$FeCl_3 + NaOH \rightarrow$ Negatively charged colloids

$FeCl_3 +$ hot water $\rightarrow$ Positively charged colloids

NUMERICAL

55. Calculate the vapor pressure of water at 300 K if its heat of vaporization is 540 cal/g.

Ans- 31.92

Sol:-

$$2.303 \log \frac{P_2}{P_1} = \frac{\Delta H}{R}\left[\frac{T_2 - T_1}{T_1 T_2}\right]$$

For water b.pt. = 373 K and thus, $P = 760$ mm;

Also, $\Delta H = 540 \times 18$ cal / mol

$$2.303 \log \frac{760}{P_1} = \frac{540 \times 18}{2} \times \left[\frac{373-300}{373 \times 300}\right]$$

$\therefore$ $P_1 = \mathbf{31.92\ mm}$

56. The variation of standard cell potential (e.m.f) of the cell,

$Pt\ H_2(g)|HBr(aq.)||AgBr(s)|Ag(s)$; at 298 K is given by;

$E^0(\text{in V}) = 0.07131 - 4.99 \times 10^{-4}T - 3.45 \times 10^{-6}T^2$

Evaluate the standard Gibb's energy change, enthalpy change and entropy change at 298 K. Assume 1F=96500C.

Ans:- 48.7

Sol:-

$$\frac{1}{2}H_2 \longrightarrow H^+ + e$$

$$Ag^+ + e \longrightarrow Ag$$

At $T = 298\text{ K}, \quad E° = +0.07131V$

$\because \quad \Delta G° = -nF \times E° = -1 \times 96500 \times 0.07131V$

$$= -6.88 \times 10^3 \text{ J} = \mathbf{-6.88\ kJ / mol}$$

$$\left(\frac{\partial E^\circ}{\partial T}\right)_P = -4.99 \times 10^{-4} - 2 \times 3.45 \times 10^{-6} T$$

$$= -5.06 \times 10^{-4}$$

Since, $\Delta H^\circ = nF\left[T\left(\frac{\partial E^\circ}{\partial T}\right)_P - E^\circ\right]$

$\therefore \quad \Delta H^\circ = 1 \times 96500[298 \times (-5.06 \times 10^{-4}) - 0.07131]$

$$= -2.14 \times 10^4 \text{ J} = \mathbf{-21.4\ kJ}$$

Also, $\quad \Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$

$\therefore \quad -\Delta S^\circ = \frac{\Delta G^\circ - \Delta H^\circ}{T}$

$$-\Delta S^\circ = \frac{-6.88 - (-21.4)}{298}$$

$$\Delta S^\circ = -0.0487 \text{ kJ} = \mathbf{-48.7\ J}$$

57. The rate constant of a certain reaction is given by.

$$\log_{10} K = A - \frac{B}{T} + C\log_{10} T$$

Prove that energy of activation for the reaction at 300K is

$$E_a = [2.303B + CT] \times R$$

Ans:- 2.303

Sol:-

$$\log_{10} K = A - \frac{B}{T} + C\log_{10} T$$

$\therefore \quad \frac{\ln K}{2.303} = A - \frac{B}{T} + \frac{C\ln T}{2.303}$

or $\quad \ln K = 2.303A - \frac{2.303B}{T} + C\ln T$

This is integration of the

or $\quad \frac{d\ln K}{dT} = \left[\frac{2.303B}{T^2} + \frac{C}{T}\right] = \frac{E_a}{RT^2}$

$\therefore \quad \frac{E_a}{RT^2} = \left[\frac{2.303B}{T^2} + \frac{CT}{T^2}\right] = \left[\frac{2.303B + CT}{T^2}\right]$

$\therefore \quad E_a = [2.303\,B + CT] \times R$

Given, $\quad H \longrightarrow H^+ + e, \qquad IE_{1_H}$

58. A ternary solution contains 2,3 and 5 mole of A,B and C respectively. Total vapour pressure of this Solution (in mm) is given by an empirical equation

$$P_T = 100X_A + 200X_B + 90X_C$$

Where, X_A, X_B and X_C are mole fractions of A,B and C in solution state. Calculate the mole fraction of A,B and C in vapour phase.

Ans:- 0.36

Sol:-

$X_A = X_B = 0$ and $X_C = 1$ then $P_C^0 = 90$ mm

$X_A = X_C = 0$ and $X_B = 1$ then $P_B^0 = 200$ mm

$X_B = X_C = 0$ and $X_A = 1$ then $P_A^0 = 100$ mm

Also, $P_T = 100 \times 0.2 + 200 \times 0.3 + 90 \times 0.5$

$= 20 + 60 + 45 = 125$ mm

using $P_T \cdot X'_A = P_A^0 \cdot X_A$

$$X'_A = \frac{100 \times 0.2}{125} = \mathbf{0.16}$$

Similarly, $$X'_B = \frac{200 \times 0.3}{125} = \mathbf{0.48}$$

$$X'_C = \frac{90 \times 0.5}{125} = \mathbf{0.36}$$

59. An aqueous solution of 0.10 N NaCl is separated from 0.10N Na_2SO_4 solution in water through a semipermeable membrane at 27^0C. what minimum pressure and on which solution should be applied to reverse the process of osmosis? Assume 100% ionization of both.

Ans:- 3.695

Sol:-

$\pi V = nST$

For NaCl : $\pi V = nST\,(1+\alpha)$ $\quad (NaCl \rightleftharpoons Na^+ + Cl^-)$

$$\pi_{NaCl} = \frac{0.1}{1} \times 0.0821 \times 300 \times 2 \quad [M_{NaCl} = N_{NaCl}]$$

$$= 4.926\,\text{atm}$$

For Na_2SO_4 : $\pi V = nST\,(1+2\alpha)$

$$(Na_2SO_4 \rightleftharpoons 2Na^+ + SO_4^{2-})$$

$$\pi_{Na_2SO_4} = \frac{0.1}{2} \times 0.0821 \times 300 \times 3$$

$$\left(M_{Na_2SO_4} = \frac{N_{Na_2SO_4}}{2}\right)$$

$$= 3.695\,\text{atm}$$

The solvent water will move from hypotonic solution (Na_2SO_4, less conc. as $M_{Na_2SO_4}$ is 0.05 *M*) to hypertonic solution (NaCl, conc. as $M_{NaCl} = 0.1M$) during osmosis. Thus, a pressure just higher than the value $(4.926 - 3.695)$, *i.e.*, **1.231 atm** must be applied on **NaCl** solution to reverse the osmosis.

60. In a parallel path reaction, each step involving elementary reactions as:

$$A \xrightarrow{K_1} 2B \qquad ....(i)$$

$$3A \xrightarrow{K_2} 4B \qquad ....(ii)$$

$$2A \xrightarrow{K_3} 2B \qquad ....(iii)$$

Frame the expression for the rate of total disappearance of A in terms of [A]

Ans:-

Sol:-

$$\text{Rate} = -\frac{d[A]}{dt} = K_1[A] \qquad \text{...from (i)}$$

$$\text{Rate} = -\frac{1}{3}\frac{d[A]}{dt} = K_2[A]^3 \qquad \text{...from (ii)}$$

$$\text{Rate} = -\frac{1}{2}\frac{d[A]}{dt} = K_3[A]^2 \qquad \text{...from (iii)}$$

For parallel path decay, the rate is additive and thus

$$\text{Total rate} = \frac{-d[A]}{dt} = \boldsymbol{K_1[A] + K_2[A]^3 + K_3[A]^2}$$

61. When an immiscible liquid with water was steam distilled at 95.2^0C at a total pressure of 747.3 torr, the distillate contained 1.27 g of liquid per gram of water. Calculate the molar mass of the liquid. The vapour pressure of water is 638.6 torr at 95.2^0C.

Ans:- 134.3

Sol:-

$P_T = 747.3$ torr; $p^1_{H_2O} = 638.6$ torr

$\therefore \quad P^1_L = 747.3 - 638.6 = 108.7$ torr

Also, $\quad \dfrac{w_L}{w_{H_2O}} = \dfrac{1.27}{1}$

Let molar mass of liquid be m_L, then according to Dalton's law of partial pressure.

$$P^1_{H_2O} = P_T \times \frac{\frac{w_{H_2O}}{18}}{\frac{w_L}{m_L} + \frac{w_{H_2O}}{18}}$$

$$P^1_L = P_T \times \frac{\frac{w_L}{m_L}}{\frac{w_L}{m_L} + \frac{w_{H_2O}}{18}}$$

$$\therefore \quad \frac{P^1_{H_2O}}{P^1_L} = \frac{w_{H_2O} \times m_L}{18 \times w_L}$$

$$\text{or} \quad \frac{638.6}{108.9} = \frac{m_L}{18 \times 1.27}$$

$$\therefore \quad M_L = 134.3$$

62. Dal lake has water 8.2×10^{12} litre approximately. A power reactor produces electricty at the rate of 1.5×10^{6} coulomb per second at an appropriate viltage. How many yeats would it take to electrolyse the lake?

Ans:- 1.9 million year

Sol :

63. A lead storage cell is discharged which causes the H_2SO_4 electrolyte to charge from a concetration of 34.6% by weight (density $1.261\,g\,ml^{-1}$ at 25^0C) to 27% by weight. The original volume of electrolyte is one litre. Caculate the total charge realesed at anode of the battery. Note that the water is produced by the cell reaction as H_2SO_4 is used up. Over all reaction is

$$Pb(s) + PbO_2(s) + 2H_2SO_4(l) \rightarrow 2PbSO_4(s) + 2H_2O(l)$$

Ans:- 1.21×10^{5} coulombs

Sol :

64. Caculate the cell potential of a cell having reaction: $Ag_2S + 2e^- \rightleftharpoons 2Ag + S^{2-}$ in a solution buffered at $pH = 3$ and which is also saturated with $0.1MH_2S$.

For $H_2S : K_1 = 10^{-8}$ and $K_2 = 1.1\times10^{-13}, K_{sp}\left(Ag_2S\right) = 2\times10^{-49}, E^0_{Ag^+/Ag} = 0.8.$

Ans: - 0.167 volts

Sol :

65. The equivalent conductance of 0.10 N solution of $MgCl_2$ is 97.1 mho $cm^2\ equi^{-1}$ at 25^0C. A cell with electrode that are $1.5cm^2$ in surface area and 0.5cm apart is filled with 0.1 N $MgCl_2$ solution. How much current will flow when potential difference between the electrodes is 5 volt.

Ans: 0.1456 Amperes

66. A dilute aqueous solution of KCl was placed between two electrodes 10 cm apart,across which a potenial of 6 volt was applied. How far would the K^+ ion move in 2 hours at 25^0C? Ionic conductance of K^+ c ion at infinite dilution at 25^0C is 73.52 $ohm^{-1}\ cm^2 mole^{-1}$?

Ans:- 3.29cm

Sol :

67. Calciulate the solubilty and solubility product of $Co_2\left[Fe(CN)_6\right]$ in water at 25^0C from the following data:

Conductivity of a sturated solution of $Co_2\left[Fe(CN)_6\right]$ is $2.06\times10^{-6}\Omega^{-1}cm^{-1}$ and that of water used $4.1\times10^{-7}\Omega^{-1}cm^{-1}$. The ionic molar conductivities of Co^{2+} and $Fe(CN)_6^{4-}$ are $86.0\Omega^{-1}\ cm^2 mol^{-1}$ and $444.0\Omega^{-1}\ cm-1\ mol^{-1}$.

Ans: $K_{SP} = 7.682\times10^{-17}$

Sol :

68. In two vessles each containing 500ml water, 0.5m mol of aniline$\left(K_b = 10^{-9}\right)$ and 25mmol of HCl are added separately. Two hydrogen electrodes are constructed using these solutions. Calculate the emf of cell made by connecting them approptiately

Ans: E = 0.395 volts

Sol:-

69. If edge fraction unoccupied in ideal anti- flurite structure is x. Calculate the value of Z. where

$$Z = \frac{x}{0.097}$$

Ans:- 3

70. Ionic acid Na^+A^- crystallise in rock salt type structure. 2.592 gm of ionic solid salf NaA dissoloved in water to make 2 litre solution. The pH of this solution is 8, If distance between cation and anion is 300 pm. Calculate density of ionic solid (in gm/cm^3).

(given: $pK_w = 13, pK_a\left(HA\right) = 5, N_A = 6\times10^{23}$)

Ans:- 4

71. Calculate the value of $\frac{Z}{10}$. Where

Z=Co-ordination number of 2D-square close packing

+

Co-ordination number of 2D-hcp

+

Co-ordination number of 3D-square close packing

+

Co-ordination number of 3D,ABCABC…..packing

+

Co-ordination number of 3D,ABAB….packing

Ans:- 4

72. Assume $KBr, Li_2O, Zinc$ blende, $Fe_2O_3, CaF_2, NaCl - all$ have perfect undistorted structures (exact fitting). How many above compounds have void fraction greater than 0.3?

Ans:- 0

73. How many square faces are there in truncated octahedron?

Ans:- 6

38. Diamond structure can be considered as ZnS (Zin blende) structure in which each Zn^{2+} in alternate tetrahedral void and S^{2-} in cubic closed pack arrangement is replaced by one carbon atom. If C-C covalent bond length in diamond is 1.5 $\mathring{A}$, what is the edge length of diamond unit cell (z=8)

Ans:- 3.46

74. A molecule X_2Y (molecular weigh=166.4) occupies orthorhombic lattice with $a = 5\mathring{A}, b = 8\mathring{A}$ and $c = 4\mathring{A}$. If density of X_2Y is 5.2 gm cm^{-3}. Calculate number of molecules present in one unit cell $(N_A = 6\times10^{23})$

Ans:- 3

75. A inverse spinel structure $(Fe_3O_4\ type)O^{-2}$ form FCC lattice, Fe^{2+} ions occupy 1/8 of the tetrahedral Voids and trivalent cation occupies 1/8 of the tetrahedral voids and ¼ of the octahedral voids. Calculate the value of Z.

Where $Z = \frac{packing\ fraction}{0.384}$.given $r_{Fe^{+2}} = 0.225 r_{O^{-2}}; r_{Fe^{+3}} = 0.414 r_{O^{-2}}$

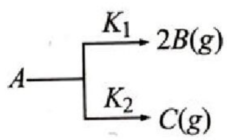
Ans:- 2

76. A metal slowly forms an oxide film which completely protects the metal when the film thickness is 3.956 thousand thus of an inch. If the film thickness is 1.281 thou. in 6 weeks, how much longer will it be before it is

2.481 thou? the rate of film formation follows first order kinetics.

Ans:- 15.13

77. Gas $A(g)$ shoes two parallel path I order reaction as:



The initial pressure of $A(g)$ in a container of volume V litre is 1 atm. After 10sec the pressure become 1.4 atm and after ∞ time it becomes 1.5 atm. Calculate K_1 and K_2.

Ans:- 0.062

Sol:-

$$\frac{-d[B]}{dt} = K_1[A]$$

$$\frac{-d[C]}{dt} = K_2[A]$$

$$\therefore \quad \frac{[B]}{[C]} = \frac{K_1}{K_2}$$

Let after ∞ time pressure equivalent to P' shows dissociation to B and $1-P'$ to C

$$\therefore \quad 2P' + 1 - P' = 1.5$$

$$\therefore \quad P' = 0.5 \text{ atm}$$

$$\therefore \quad \frac{K_1}{K_2} = \frac{2P'}{1-P'} = \frac{1}{0.5} = 2$$

i.e., 2 : 1 is the ratio for the decomposition to $B : C$.

Now,

$$\begin{array}{cccc} A & \longrightarrow & 2B & + \quad C \\ 1 & & 0 & \quad 0 \\ 1-a & & \frac{2\times 2a}{3} & \quad \frac{1\times a}{3} \end{array}$$

$$\therefore \quad 1 - a + \frac{4a}{3} + \frac{a}{3} = 1.4 \qquad \therefore \; a = 0.6$$

Also, $$K_{av} = \frac{2303}{10}\log\frac{1}{0.4} = 0.092$$

$$K_{av} = K_1 + K_2$$

$$0.092 = 2K_2 + K_2$$

$$\therefore \quad K_2 = 0.03; \quad K_1 = 0.062$$

78. A first order reaction $A \rightarrow B$ requires activation energy of 70 kJ mol^{-1}. When a 20% solution of A was kept at $25^0 C$ for 20 minute, 25% decomposition took place. What will be the per cent decomposition in the same time in a 30% solution maintained at $40^0 C$? Assume that activation energy remain

constant in this range of temperature.

Ans:- 67.21

Sol:-

Given, $A \longrightarrow B$

and 20% solution of A decomposes 25% in 20 minute at 25°C

$\therefore$ Initial amount $a = 20$

$\therefore$ Amount left $(a-x) = 20 \times \frac{75}{100} = 15$

$\therefore$ $K_{25} = \frac{2.303}{20}\log_{10}\frac{20}{15}$ $(\because \ t = 20 \text{ minute})$

$= 0.0144 \text{ minute}^{-1}$

Now, $2.303\log_{10}\frac{K_{40}}{K_{25}} = \frac{E_a}{R}\frac{[T_2 - T_1]}{T_1T_2}$

$\therefore$ $2.303\log_{10}\frac{K_{40}}{K_{25}} = \frac{70\times10^3}{8.314}\left[\frac{313-298}{313\times298}\right]$

$\frac{K_{40}}{K_{25}} = 3.872$

$\therefore$ $K_{40} = 0.05575 \text{ min}^{-1}$

$(\because \ K_{25} = 0.0144)$

Now suppose amount m is left in 30% solution in 20 minute at 40°C.

$K_{40} = \frac{2.303}{t}\log_{10}\frac{a}{(a-x)}$

$0.05575 = \frac{2.303}{20}\log_{10}\frac{30}{m}$

$\therefore$ $m = 9.838$

$\therefore$ % decomposition $= \frac{(a-m)}{a}\times100$

$= \frac{30-9.838}{30}\times100 = \mathbf{67.21\%}$

79. A given sample of milk turns sour at room temperature $\left(20^0C\right)$ in 64 hour. In a refrigerator at 3^0C milk can be stored three times as long before it sours. Estimate (a) the activation energy for souring of milk, (b) how long it take milk to sour at 40^0C?

Ans:- 276

Sol:- Given, $\frac{K_{293}}{K_{276}} = 3; T_2 = 293K$ and $T_1 = 276K$

(a) $2.303\log_{10}\frac{K_2}{K_1}=\frac{E_a}{R}\frac{[T_2-T_1]}{T_1T_2}$ $(R=2\,\text{cal})$

$\therefore$ $2.303\log_{10}3=\frac{E_a}{2}\frac{[293-276]}{293\times276}$

$\therefore$ $E_a=10453.95\,\text{cal}=\mathbf{10.454\ kcal}$

(b) Also, $2.303\log_{10}\frac{K_3}{K_2}=\frac{E_a}{R}\frac{[T_3-T_2]}{T_3T_2}$

This time, $E_a=10.454$ kcal; $T_3=313$ K and $T_2=293$ K

$\therefore$ $2.303\log_{10}\frac{K_3}{K_2}=\frac{10.454\times10^3}{2}\frac{[313-293]}{313\times293}$

$\therefore$ $\frac{K_3}{K_2}=3.12$

Now, $\frac{K_3}{K_2}=\frac{t_2}{t_3}$ $\because$ $K\propto\frac{1}{\text{time}}$

Also if milk is not soured up to 64 hr at 20°C, it will not sour up to 192 hr at 3°C. Similarly we can have

$$t_3=t_2\times\frac{K_2}{K_3}=64\times\frac{1}{3.12}=\mathbf{20.5\ hr}$$

80. For the reaction, $2NO+H_2\rightarrow N_2O+H_2$, the value of $-\frac{dP}{dt}$ was found to be 1.5 $Pa\,s^{-1}$ for a pressure of 359 Pa of NO and 0.25 Pa s^{-1} for a pressure of 152 Pa of NO. The pressure of H_2 being constant. If pressure of NO was kept constant, the value of $-\frac{dP}{dt}$ was found 1.60 Pa s^{-1} for a pressure of H_2 289 Pa and 0.79 Pa s^{-1} for a pressure of 147 Pa of H_2. Calculate the order of reaction.

Ans:- 3

Sol:-

Rate of reaction $=-\frac{1}{2}\frac{dP_{NO}}{dt}=-\frac{dP_{H_2}}{dt}$

For P_{H_2} constant :

$$-\frac{1}{2}\frac{dP_{NO}}{dt}=1.5=K(359)^m(P_{H_2})^n$$

$$-\frac{1}{2}\frac{dP_{NO}}{dt}=0.25=K(152)^m(P_{H_2})^n$$

$\therefore$ $\frac{1.5}{0.25}=\left(\frac{359}{152}\right)^m$ $\therefore$ $\mathbf{m=2}$

For P_{NO} constant :

$$-\frac{dP_{H_2}}{dt} = 1.60 = (P_{NO})^m (289)^n$$

$$-\frac{dP_{H_2}}{dt} = 0.79 = (P_{NO})^m (147)^n$$

$$\therefore \quad \frac{1.60}{0.79} = \left(\frac{289}{147}\right)^n \quad \therefore \; n = 1$$

$$\therefore \quad \text{Rate} = K[P_{NO}]^2[P_{H_2}]^1$$

$\therefore$ Order of Reaction $= 2+1 = 3$

81. A certain reaction B^{n+} is getting converted to $B^{(n+4)+}$ in solution. The rate constant of this reaction is measured by titrating a volume of the solution with a reducing agent which reacts only with B^{n+} and $B^{(n+4)+}$. In this process it converts B^{n+} to $B^{(n+2)+}$ and $B^{(n+4)+}$ to $B^{(n-1)+}$. At t=0, the volume of reagent consumed is 25 mL and at t=10 minute. The volume used is 32 mL. Calculate the rate constant for the conversion of B^{n+} of $B^{(n+4)+}$ assuming it to be a first order reaction.

Ans:-

Sol:-

$$B^{n+} \longrightarrow B^{(n+4)+}$$

Millimole at $t=0$	a	0	$2e + B^{n+} \longrightarrow B^{(n+2)+}$
$t=t$	$(a-x)$	x	$5e + B^{(n+4)+} \longrightarrow B^{(n-1)+}$

Let normality be N for reducing agent.

Thus, at $t=0$ $\quad a \times 2 = N \times 25 \;\therefore\; a = \frac{25}{2}N$

at $t=t$ $\quad \underset{\text{For } B^{n+}}{(a-x)\times 2} + \underset{\text{for } B^{(n+4)+}}{x \cdot 5} = N \times 32$

$$\therefore \quad 3x = 7N \quad \text{or} \quad x = \frac{7}{3}N$$

$$\text{Now,} \quad K = \frac{2.303}{10}\log\frac{25/2N}{\left(\frac{25}{2}-\frac{7}{3}\right)N} = \frac{2.303}{10}\log\frac{25\times 6}{2\times 61}$$

$$K = \mathbf{2.07 \times 10^{-2}\ min^{-1}}$$

82. For reversible first order reaction,

$$A \underset{K_b}{\overset{K_f}{\rightleftharpoons}} B; K_f = 10^{-2}\ \text{sec}^{-1}$$

And $\frac{B_{eq}}{A_{eq}} = 4$; If $A_0 = 0.01ML^{-1}$ and $B_0 = 0$, what will be concentration of B and 30 sec?

Ans:-

Sol:-

$$A_0 \longrightarrow B$$
$$0.01 \qquad 0$$
$$0.01 - x_{eq.} \qquad x_{eq.}$$
$$\frac{[B]_{eq.}}{[A]_{eq.}} = \frac{10^{-2}}{K_b} = 4 = \frac{[x]_{eq.}}{0.01-[x]_{eq.}}$$

$\therefore$ $$K_b = 0.25\times10^{-2}$$

and $$x_{eq.} = \frac{0.04}{5} = 0.008$$

$$t = \frac{2.303}{(K_f + K_b)} \log\frac{[x]_{eq.}}{[x]_{eq.} - x}$$

$$30 = \frac{2.303}{1.25\times10^{-2}} \log\frac{0.008}{(0.008-x)}$$

$\therefore$ $$\frac{0.008}{0.008-x} = 1.455$$

$\therefore$ $$x = \mathbf{2.50\times10^{-3}}$$

COMPREHENSION

When sucrose is hydrolyzed, it produced dextrorotatory "glucose" and leavorotatory "fructose" whereas the reactant itself is a dextrorotatory compound. The above conversion process follows first order kinetics and the process is called inversion of cane sugar. Similarly an optically active compound A is hydrolysed as follows.

$$A + H_2O \xrightarrow{H^+} B + C$$

The observed rotation of compounds A,B and C are 60^0, 50^0 and 80^0 per mole respectively. The angles of rotation after 40 minute and after the completion of reaction were 26^0 and 10^0 respectively, at 27^0C and activation energy for conversion is $27kJ\ mol^{-1}$.

83. The value of rate constant of the above reaction at 27^0C is

A) $5.58\times10^{-3}\ min^{-1}$ B) $1.94\times10^{-3}\ min^{-1}$ C) $1.37\times10^{-6}\ min^{-1}$ D) $6.13\times10^{-3}\ min^{-1}$

Key:- A

Sol:-

(a) $K = \frac{2.303}{t} \log \frac{a}{(a-x)}$;

$r_0 = 60 \propto$ (Sugar + ext. factor)

$r_t = 26 \propto$ [Sugar left + ext. factor + difference in rotation due to B and $C(30)$]

$r_\infty = 10 \propto$ [ext. factor + difference in rotation due to B and $C(30)$]

$(60-10) \propto$ Sugar + (30)

$\therefore$ Sugar $= a \propto 20$

Similarly, $26-10 \propto$ Sugar left

$\therefore$ Sugar left $= a - x \propto 16$

$\therefore$ $K = \frac{2.303}{40} \log \frac{20}{16} = \mathbf{5.58 \times 10^{-3}\ min^{-1}}$

84. The value of $t_{1/2}$ for above process at 127^0C is

A) 5.68 minute B) 1.2 hour C) 8.298 hour D) 1.5 hour

Key:- C

Sol:-

$$2.303 \log \frac{K_2}{K_1} = \frac{\Delta H}{R} \frac{[T_2 - T_1]}{T_1 T_2}$$

$$2.303 \log \frac{K_2}{5.58 \times 10^{-3}} = \frac{27}{8.314 \times 10^{-3}} \times \frac{100}{300 \times 400}$$

$$K_2 = 0.0835 \text{ at } 127^\circ \text{C}$$

$$\therefore \quad t_{1/2} = \frac{0.693}{0.0835} = \mathbf{8.298\ min}$$

85. Rate of formation of B if rate of reaction at 27^0C is $r = K[A]^1[H^+]^0$ and $[A] = 0.2M/L$ is

A) $2.23 \times 10^{-3} ML^{-1} min^{-1}$ B) $1.11 \times 10^{-3} ML^{-1} min^{-1}$

C) $5.58 \times 10^{-3} ML^{-1} min^{-1}$ D) $0.58 \times 10^{-3} ML^{-1} min^{-1}$

Key:- B

Sol:- $\frac{d[B]}{dt} = -\frac{d[A]}{dt}$

86. The reaction $A + H_2O \xrightarrow{H^+} B + C$, is called

A) Acid-base catalyzed reaction B) mutarotation if A is glucose

C) inversion of cane sugar if A is sucrose D) either of the above

Key:- D

Sol:- All are facts

COMPREHENSION:- 7 QUESTIONS.

Only a few reactions are simple or straight forward, i.e., purely first, second or third order Reactions. But in most of the reaction, the interpretation of variety data becomes difficult due to several other reactions taking place simultaneously along with the main reaction and thus smooth progress of the reaction is disturbed. Such reactions includes the following different types. The reactions in which a substance reacts or decomposes in more than one way, are called parallel or competing or side reaction. The reactions which take place exclusively on the walls of containing vessels are known as surface reactions. The reactions in which the products of chemical change react together to form the original reactants are called reversible reactions or opposing reactions or counter reactions. The reactions which proceeds from reactants to final products through one or more intermediate stages are called consecutive reactions

87. The relation $\Delta H = E_{a(f)} - E_{a(b)}$ is valid for

A) side reactions
B) counter reactions
C) Surface reactions
D) consecutive reaction

Key:- B

Sol:- Forward and backward reactions occur only in reversible reactions.

88. Radioactive decay of ^{235}U to emit $\alpha-$ and $\beta-$particles includes.

(i) parallel reactions (ii) counter reactions (iii) surface reactions
(iv) consecutive reactions

A) i,ii,iii B) ii,iii C) i,iv D) ii,iv

Key:- C

Sol:- The decay involves consecutive reaction as well as some intermediates decay to give parallel Reactions.

89. Formation of o, m and p-nitrobenzoic acid during nitration of nitrobenzene involves.

A) Competing reactions with main reaction giving o-and p-nitrobenzoic acid
B) Competing reactions with main reaction products as m-nitrobenzoic acid
C) Suraface reactions with main reaction products as m-nitrobenzoic acid
D) Consecutive reactions with main reaction product as o and p-nitrobenzoic acid

Key:- B

Sol:- This is an example of parallel path decay forming m-nitrobenzoic acid as- NO_2 is m-directing

90. The ratio of amounts of substances formed in two side reactions is.

A) Independent of time, provided the order of two side reactions is same

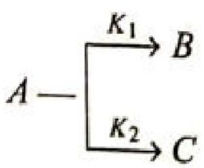
B) Dependent of time provided the order of two side reactions is same

C) Decreases with time provided the order of two side reactions is same

D) Increases with time provided the order of two side reactions is same

Key:- A

Sol:-



$$\frac{\text{amount of B at any stage}}{\text{amount of C at any stage}} = \frac{\text{rate of formation of B}}{\text{rate of formation of C}}$$

$$= \frac{K_1(a-X)^n}{K_2(a-X)^n} = \frac{K_1}{K_2}$$

91. The study of the kinetics of….. has led to the discovery that in addition to reactions of simple orders, there are other reactions which are fractional order or negative orders:

A) Side reactions B) surface reactions

C) reversible reactions D) Consecutive reactions

Key:- B

Sol:- This is a fact.

92. The decomposition of NH_3 on platinum surface takes place following rate law

$$\frac{dx}{dt} = K[NH_3][H_2]^{-1}$$

This is an example of:

A) side reaction B) surface reaction C) counter reaction D)consecutive reaction

Key:- B

Sol:- It is a fact

93. Oxidation of H_3PO_3 by $K_2S_2O_8$ in presence of HI takes place in two steps as

$$K_2S_2O_8 + 2HI \longrightarrow H_2SO_4 + K_2SO_4 + I_2 \text{ (fast)}$$

$$H_3PO_4 + I_2 + H_2O \longrightarrow H_3PO_4 + 2HI \text{ (Slow)}$$

This is an example of

A) side reaction B) surface reaction C) consecutive reaction D) reversible reaction

Key:- C

Sol:- It is a fact

COMPREHENSION:- 4 QUESTIONS.

The liquid junction potential or diffusion potential arises mainly due to unequal rates of diffusion of ions across the junction (interface), i.e., it is due to the difference in migration of ions. Because different ions have different speeds, they tends to diffuse across the boundary at different speeds for example in cell $Pt_{H_2}(g)\left|\begin{array}{c:c} HCl & HCl \\ I & II \\ C_1 & C_2 \end{array}\right| H_2Pt(g)$, both H^+ and Cl^- ions diffuse from concentrated solution to dilute solution. The ionic speed of H^+ is greater than Cl^-, so H^+ will diffuse more readily from I to II $(C_1 > C_2)$. The dilute solution in II compartment will thus become positively charged with respect to solution I. Consequently solution I is left with excess of Cl^- ions and thus acquires negative charge. As a result of this a double layer of +ve and –ve charges is formed at the junction of two solutions and a potential difference is set up at the junction. This is called liquid junction potential. A salt having two ions of nearly same speed (e.g., KCl, NH_4NO_3) are used as salt bridge placed between two electrodes which minimizes or eliminates liquid junction potential.

94. Which of the following are correct about liquid junction potential?

(i) The liquid junction potential can be measured directly

(ii) The liquid junction potential is due to different ionic mobility of ions

(iii)_ the magnitude of liquid junction potential depends upon relative speed of ions

(iv) the liquid junction potential always increases the e.m.f. of cell

A) i,ii,iii B) ii,iii,iv C) i,iv D) ii,iii

Key:- D

Sol:- The liquid junctions potential decreases the e.m.f. of cell ad cannot be measured directly. It id calculated.

95. Select the correct statements about liquid junction potential:

(i) The sign and magnitude of liquid junction potential depends upon transport number of anions and cations $(t_a \text{ and } t_c)$

(ii) If $t_a = t_c, E_{ljp} = 0$

(iii) If $t_a > t_c, E_{ljp}$ increases the e.m.f of cell and if $t_a < t_c, E_{ljp}$ decreases the e.m.f. of cell

(iv) Only two vertical lines between two electrodes of liquid junction potential

A) i,ii,iii B) i,iv C) ii,iii,iv D) i, iii, iv

Key:- A

Sol:- Only one vertical line (|) indicates liquid junction potential. Two vertical lines (||) indicates salf Bridge i.e., no liquid junction potential.

96. In which of the following cells, liquid junction potential does not interferes with the exact , Measurement of cell e.m.f?

A) $Zn|ZnSO_{4_{Sol}}||CuSO_4|Cu$

B) $Pt\,H_2(g)\left|\begin{matrix}HCl\\C_1\end{matrix}\right|\left|\begin{matrix}HCl\\C_2\end{matrix}\right|Pt\ H_2(g)$

C) $Pt\,Cl_2(g)\left|\begin{matrix}KCl\\C_1\end{matrix}\right|\left|\begin{matrix}KCl\\C_2\end{matrix}\right|Pt\,Cl_2(g)$

D) $Ag|AgCl(s),\underset{a_1}{HCl}||\underset{a_2}{HCl},AgCl(s)|Ag$

Key:- B

Sol:- Follow answer 2. Also, KCl has same speef of K^+ and Cl^-

97. Select the incorrect sentence

A) E.m.f of concentration cells with transport includes liquid junction potential

B) E.m.f. of concentration cells with transport does not include liquid junction potential.

C) E.m.f. of concentration cells without transport does not include liquid junction potential

D) E.m.f of liquid junction potential can be evaluated by using concentration cells with and without transport with same electrodes and same electrolyte concentrations.

Key:- B

Sol:- Rest all are facts

COMPREHENSION:- 3 QUESTIONS.

The gaseous reaction: $n_1A(g)\rightarrow n_2B(g)$ is first order with respect to A. The true rate constant of reaction is k.The reaction is studies at a constant pressure and temperature. Initially, the moles of A were 'a' and no B were present.

98. How many moles of A are present at time, t?

A) $a\cdot e^{-kt}$ B) $a\cdot e^{-n_1kt}$ C) $a\cdot e^{-n_2kt}$ D) $a\left(1-e^{-n_1kt}\right)$

Key:- B

99. If the initial volume of system were v_0, then the volume of system after time, t, will be

A) $\frac{n_1v_0}{n_2}$ B) $\frac{n_2v_0}{n_1}$

C) $v_0\left[\frac{n_2}{n_1}-\left(1-\frac{n_2}{n_1}\right).e^{-n_1kt}\right]$ D) $v_0\left[\frac{n_2}{n_1}-\left(\frac{n_2}{n_1}-1\right).e^{-n_1kt}\right]$

Key:- D

100. What will be the concentration of A at time t, if $n_1=1$ and $n_2=2$?

A) $[A_0]\cdot e^{-kt}$ B) $[A_0]\left(\frac{e^{-kt}}{2-e^{-kt}}\right)$ C) $[A_0]\left(\frac{e^{-kt}}{1-e^{-kt}}\right)$ D) $[A_0](1-2\cdot e^{-kt})$

Key:- B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